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HOME ELECTRICITY COST ESTIMATOR
TEST-CASE LOG
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TEST CASE 1: Valid Billing Days
Input: 31
Expected Result: Accepts 31 billing days.
Actual Result: PASSED.
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HOME ELECTRICITY COST ESTIMATOR
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Enter billing days (1-31): 31

TEST CASE 2: Invalid Billing Days
Input: abc, 0, 32, then 30
Expected Result: Rejects invalid values and asks again.
Actual Result: PASSED.
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HOME ELECTRICITY COST ESTIMATOR
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Enter billing days (1-31): abc
Invalid input. Enter a whole number from 1 to 31.
Enter billing days (1-31): 0
Invalid input. Enter a whole number from 1 to 31.
Enter billing days (1-31): 32
Invalid input. Enter a whole number from 1 to 31.
Enter billing days (1-31): 30

========== MAIN MENU ==========
1. Add Appliance
2. View Appliances
3. Calculate Electricity Cost
4. Search Appliance
5. Update Appliance
6. Remove Appliance
7. Find Highest Cost
8. Update Electricity Rate
9. Update Monthly Budget
10. Exit
Choose an option (1-10):

TEST CASE 3: Add an Appliance
Input:
Name: Electric Fan
Wattage: 75
Hours per Day: 8
Expected Result: Saves the appliance information.
Actual Result: PASSED.
--- ADD APPLIANCE ---
Enter appliance name: Electric Fan
Enter wattage in watts (1-10000): 75
Enter hours used per day (0.01-24): 8
Appliance added successfully!

TEST CASE 4: Invalid Appliance Input
Input: Blank name, 0 watts, 10001 watts, or 25 hours.
Expected Result: Rejects invalid values and asks again.
Actual Result: PASSED.
--- ADD APPLIANCE ---
Enter appliance name:
Input cannot be empty. Please try again.
Enter appliance name: Aircon
Enter wattage in watts (1-10000): 10001
Invalid input. Enter a whole number from 1 to 10000.
Enter wattage in watts (1-10000): 0
Invalid input. Enter a whole number from 1 to 10000.
Enter wattage in watts (1-10000): 10000
Enter hours used per day (0.01-24): 25
Invalid input. Enter a number from 0.01 to 24.
Enter hours used per day (0.01-24): 24
Appliance added successfully!

TEST CASE 5: View Appliances
Action: Select menu option 2.
Expected Result: Displays saved appliance records.
Actual Result: PASSED.
--- APPLIANCE LIST ---

No.	Name	Watts	Hours/Day
1	Electric Fan	75	8.00
2	Aircon	10000	24.00

TEST CASE 6: Calculate Electricity Cost

Input:
Wattage: 75 watts
Hours per Day: 8
Billing Days: 30
Electricity Rate: PHP 12.1188 per kWh

Expected Result:
Daily Consumption: 0.6 kWh
Monthly Consumption: 18 kWh
Estimated Cost: PHP 218.14

Actual Result: PASSED.
--- ELECTRICITY COST ESTIMATE ---
Appliance Daily kWh Monthly kWh Cost

Electric Fan 0.60 18.00 PHP 218.14

Billing days: 30
Electricity rate: PHP 12.1188 per kWh
Total electricity usage: 18.00 kWh
Estimated total cost: PHP 218.14
Status: WITHIN BUDGET.
Remaining budget: PHP 7781.86
Average daily cost: PHP 7.27
Extra days after full weeks: 2
A valid appliance list and billing period were used.
Calculation completed.

TEST CASE 7: Invalid Menu Selection
Input: abc or a number outside 1-10.
Expected Result: Rejects invalid menu choices.
Actual Result: PASSED.
Choose an option (1-10): abc
Invalid input. Enter a whole number from 1 to 10.
Choose an option (1-10): 11
Invalid input. Enter a whole number from 1 to 10.
Choose an option (1-10):

TEST CASE 8: Exception Handling
Action: Test an appropriate exception using a controlled test case.
Expected Result: The appropriate catch block handles the exception and displays an error message.
Actual Result: PASSED.

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 HOME ELECTRICITY COST ESTIMATOR
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Unexpected error: Test exception handling

Program session ended.
PS E:\Code Files\IT301\C#>

TEST CASE 9: Exit Program
Action: Select menu option 10.
Expected Result: Ends the menu loop and displays
"Program session ended."
Actual PASSED.
========== MAIN MENU ==========
1. Add Appliance
2. View Appliances
3. Calculate Electricity Cost
4. Search Appliance
5. Update Appliance
6. Remove Appliance
7. Find Highest Cost
8. Update Electricity Rate
9. Update Monthly Budget
10. Exit
Choose an option (1-10): 10
Thank you for using the program!

Program session ended.

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TESTING REMINDER:
Update the actual results and status after running
each test. Mark a test as Passed only if the observed
result matches the expected result.

// Note: Menu options 4 to 9 are not implemented.
// and not the priority for the Activity
// just for extra temporary menu list that can implement soon as the project approved.
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*/